

Chapter 7

Right Triangles and Trigonometry

Unit 2 — Triangles

McGraw Hill Glencoe Geometry Texas, 2015

Key Vocabulary

Term	Definition
Hypotenuse	Side opposite the right angle (longest side)
Legs	The two sides that form the right angle
Geometric mean	x such that $\frac{a}{x} = \frac{x}{b}$, so $x = \sqrt{ab}$
Trigonometric ratio	Ratio of two sides of a right triangle
Angle of elevation	Angle above the horizontal
Angle of depression	Angle below the horizontal

Theorem 7-1: Pythagorean Theorem

In a right triangle, the square of the **hypotenuse** equals the sum of squares of the **legs**.

$$a^2 + b^2 = c^2$$

where c is the hypotenuse and a, b are the legs.

Example: Legs 3 and 4 $\rightarrow$ Hypotenuse $= \sqrt{9 + 16} = \sqrt{25} = 5$.

Theorem 7-2: Converse of Pythagorean Theorem

If $a^2 + b^2 = c^2$, then the triangle is a **right triangle**.

Condition	Triangle type
$a^2 + b^2 = c^2$	Right
$a^2 + b^2 > c^2$	Acute
$a^2 + b^2 < c^2$	Obtuse

Pythagorean Triples

Common integer sets satisfying $a^2 + b^2 = c^2$:

Triple	Multiples
3, 4, 5	6-8-10, 9-12-15, ...
5, 12, 13	10-24-26, ...
8, 15, 17	—
7, 24, 25	—

Theorem 7-3: Geometric Mean (Altitude)

In a right triangle, the **altitude** to the hypotenuse is the geometric mean between the two segments of the hypotenuse.

$$\frac{AD}{CD} = \frac{CD}{DB} \Rightarrow CD = \sqrt{AD \cdot DB}$$

Theorem 7-4: Geometric Mean (Legs)

Each **leg** is the geometric mean of the hypotenuse and the segment of the hypotenuse adjacent to that leg.

$$\frac{AB}{AC} = \frac{AC}{AD} \Rightarrow AC = \sqrt{AB \cdot AD}$$

Special Right Triangles

45°–45°–90° Triangle

$$\text{legs} = x \qquad \text{hypotenuse} = x\sqrt{2}$$

Example: Leg = 5 $\Rightarrow$ hypotenuse = $5\sqrt{2}$

Special Right Triangles

30°–60°–90° Triangle

$$\text{short leg} = x \quad \text{long leg} = x\sqrt{3} \quad \text{hypotenuse} = 2x$$

Angle	Opposite side
30°	x (short leg)
60°	$x\sqrt{3}$ (long leg)
90°	$2x$ (hypotenuse)

Example: Short leg = 4 $\Rightarrow$ Long leg = $4\sqrt{3}$, Hyp = 8

Trigonometric Ratios

For a right triangle with acute angle θ :

$$\sin \theta = \frac{\text{opposite}}{\text{hypotenuse}}$$

$$\cos \theta = \frac{\text{adjacent}}{\text{hypotenuse}}$$

$$\tan \theta = \frac{\text{opposite}}{\text{adjacent}}$$

Memory aid: SOH-CAH-TOA

Ratio	Abbr.	Meaning
Sine	SOH	Opposite/ H ypotenuse
Cosine	CAH	A djacent/ H ypotenuse
Tangent	TOA	Opposite/ A djacent

Common Trig Values

θ	$\sin \theta$	$\cos \theta$	$\tan \theta$
30°	$\frac{1}{2}$	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{3}}{3}$
45°	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{2}}{2}$	1
60°	$\frac{\sqrt{3}}{2}$	$\frac{1}{2}$	$\sqrt{3}$

Example — Finding a Side

In right $\triangle ABC$, $\angle B = 90^\circ$, $\angle A = 35^\circ$, $AB = 10$.

Find BC :

$$\tan 35^\circ = \frac{BC}{AB} = \frac{BC}{10}$$

$$BC = 10 \cdot \tan 35^\circ \approx 10 \times 0.700 = 7.00$$

Example — Finding an Angle

In right $\triangle ABC$, $\angle B = 90^\circ$, $AB = 6$, $AC = 10$.

$$\cos A = \frac{AB}{AC} = \frac{6}{10} = 0.6$$

$$\angle A = \cos^{-1}(0.6) \approx 53.1^\circ$$

Angle of Elevation and Depression

- **Angle of elevation:** measured upward from horizontal
- **Angle of depression:** measured downward from horizontal

They are **alternate interior angles** when the horizontal lines are parallel → equal measures.

$$\theta_{\text{elevation}} = \theta_{\text{depression}}$$

Chapter 7 — Theorem Summary

#	Name	Key Idea
Thm 7-1	Pythagorean Theorem	$a^2 + b^2 = c^2$
Thm 7-2	Converse Pythagorean	$a^2 + b^2 = c^2 \rightarrow \text{right } \triangle$
Thm 7-3	Geo. Mean (Altitude)	Altitude = geo. mean of hyp. segments
Thm 7-4	Geo. Mean (Leg)	Leg = geo. mean of hyp. and adj. segment
—	45-45-90	Legs x ; hyp $x\sqrt{2}$
—	30-60-90	Sides $x, x\sqrt{3}, 2x$
—	SOH-CAH-TOA	$\sin, \cos, \tan$ ratios